Supplementary Table 1. Description of simulation scenarios used in evaluations of the IV-MVB method and comparisons to existing methods.

Scenario	Description	Parameter Settings
A	Null causal effect, moderate sample size, and varying instrument strength.	$\beta_1 = 0$ $n = 1000$ α_1 varying from 0.01 to 1.0 $p_Z = 0.2$, $\alpha_0 = 0$, $\beta_0 = 0$, $\sigma_U = 0.5$, $c_X = 1.5$, $c_Y = 1.5$, $1 - \alpha = 0.95$, $n_{\text{sims}} = 8000$
B	Positive causal effect, moderate sample size, and varying instrument strength.	$\beta_1 = 1$ $n = 1000$ α_1 varying from 0.01 to 1.0 $p_Z = 0.2$, $\alpha_0 = 0$, $\beta_0 = 0$, $\sigma_U = 0.5$, $c_X = 1.5$, $c_Y = 1.5$, $1 - \alpha = 0.95$, $n_{\text{sims}} = 8000$
C	Null causal effect, varying sample size, and moderate instrument strength.	$\beta_1 = 0$ n varying from 50 to 2500 $\alpha_1 = 0.7$ $p_Z = 0.2$, $\alpha_0 = 0$, $\beta_0 = 0$, $\sigma_U = 0.5$, $c_X = 1.5$, $c_Y = 1.5$, $1 - \alpha = 0.95$, $n_{\text{sims}} = 8000$
D	Positive causal effect, varying sample size, and moderate instrument strength.	$\beta_1 = 1$ n varying from 50 to 2500 $\alpha_1 = 0.7$ $p_Z = 0.2$, $\alpha_0 = 0$, $\beta_0 = 0$, $\sigma_U = 0.5$, $c_X = 1.5$, $c_Y = 1.5$, $1 - \alpha = 0.95$, $n_{\text{sims}} = 8000$